

HANA AL HARIS

AI/ML Engineer • GenAI & Agentic Systems • Data Science • FinTech AI

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PROFESSIONAL SUMMARY

Final-year Computer Science (AI & Machine Learning) student with hands-on experience across the full AI/ML spectrum—Generative AI and agentic systems, predictive machine learning, causal inference, and production MLOps. Proficient in Python, LangChain, LangGraph, Scikit-learn, XGBoost, FastAPI, Docker, and MLflow. Built and deployed end-to-end AI systems spanning financial risk detection, LLM safety, and customer analytics — architecting, developing, and shipping independently from concept to production.

TECHNICAL SKILLS

Languages: Python (primary), SQL, Bash

Generative AI & Agentic: LangChain, LangGraph, RAG Pipelines, Multi-Agent Orchestration, Tool-Use Agents, Memory & Planning Loops, Prompt Engineering, FAISS, ChromaDB

ML & Predictive: Scikit-learn, XGBoost, Isolation Forest, LOF, One-Class SVM, GraphSAGE GNN, Ensemble Methods, Feature Engineering (80+ features)

Causal Inference: DoWhy, EconML, DAGs, Backdoor Criterion, CATE Estimation, T-Learner, S-Learner, Uplift Modelling

LLM Safety & Governance: Prompt Injection Detection, Jailbreak Detection, PII Redaction (Microsoft Presidio), RAG Grounding, Hallucination Detection, YAML Policy Engines

MLOps & Deployment: MLflow (experiment tracking + registry), FastAPI, Docker, Docker Compose, GitHub Actions CI/CD, AWS EC2, Drift Detection (PSI, KS Test)

Data Engineering: Pandas, NumPy, SciPy, SQLAlchemy, Data Validation, Feature Drift Monitoring

Visualisation & BI: Streamlit, Plotly, Power BI, Interactive Dashboards

Databases: MySQL, PostgreSQL, MongoDB

Tools: Git, GitHub, VS Code, Jupyter, Linux

KEY PROJECTS

Financial Risk Intelligence Platform

2026

Tech: Python, LangChain, LangGraph, XGBoost, GraphSAGE, FAISS, ChromaDB, FastAPI, Docker, MLflow, GitHub Actions

- Architected a two-sub-system financial AI platform—`finrisk_detection_engine` (multi-layer AML detection ML pipeline) and `finintel_sentinel_ai` (agentic RAG investigation system)—covering autonomous financial crime detection and compliance investigation end-to-end
- `finrisk_detection_engine`: production ML pipeline with rule-based AML engines, statistical anomaly detectors (Z-Score, IQR, MAD), XGBoost ensemble models, and GraphSAGE GNN fraud ring detection—real-time FastAPI scoring API with MLflow experiment tracking, alert severity classification, and feature drift monitoring using PSI and KS tests
- Deployed the containerised scoring service to an AWS EC2 instance, exposing the FastAPI inference API over a public endpoint—first hands-on cloud deployment experience beyond local Docker environments
- `finintel_sentinel_ai`: built a production RAG pipeline—document ingestion, chunking, embedding, and vector storage using FAISS and ChromaDB—multi-agent LangGraph orchestration with tool-use, memory, and autonomous planning loops; generates structured, grounded outputs with confidence scoring and human-in-the-loop feedback
- Owned the full system end-to-end—architecture, model development, API deployment, Docker containerisation, and CI/CD via GitHub Actions—built and shipped independently

FinGuard — LLM Guardrails Gateway

2026

Tech: Python, FastAPI, LangChain, Microsoft Presidio, Pydantic, PyYAML, Docker, GitHub Actions

- Built a production LLM security middleware layer—validates both input and output on every LLM call—deployed as a Dockerized FastAPI microservice with CI/CD via GitHub Actions
- Input guardrails: prompt injection detection, jailbreak detection, and PII redaction using Microsoft Presidio; output guardrails: Pydantic schema validation, RAG grounding checks, hallucination detection, and off-topic response flagging

- Policy engine: YAML-driven rules—non-engineers can update guardrail configurations without code changes, enabling compliance teams to own AI quality standards independently

Customer Engagement & Retention Analytics — Causal Churn Intelligence

2026

Tech: Python, Pandas, Scikit-learn, DoWhy, EconML, Streamlit, Plotly, SQL

- Engineered five novel KPIs (Engagement Retention Ratio, Product Depth Index, High-Balance Disengagement Rate, Credit Card Stickiness Score, Relationship Strength Index) through EDA and statistical analysis across customer engagement dimensions
- Applied causal inference—constructed a DAG using DoWhy to identify confounders (Age, Balance, Geography) and estimated the causal effect of product ownership on churn using backdoor criterion; validated with refutation tests
- Estimated Conditional Average Treatment Effects (CATE) using EconML meta-learners (T-Learner, S-Learner) to produce a ranked uplift model for cross-sell and retention strategy; built an interactive Streamlit dashboard with a causal intervention panel

Business Anomaly Intelligence Platform

2025 – 2026

Tech: Python, Scikit-learn, FastAPI, Docker, MLflow, SciPy, Pandas, NumPy, GitHub Actions

- Architected an 11-stage production ML pipeline—extraction, validation, feature engineering (80+ features), statistical and ML ensemble detection, drift monitoring, and alert severity classification—a full end-to-end, reproducible data science workflow
- Built a 3-model anomaly detection ensemble (Isolation Forest, LOF, One-Class SVM) with pairwise agreement scoring and ensemble consensus voting; containerised with Docker + docker-compose, FastAPI inference service with CI/CD via GitHub Actions

EXPERIENCE

AI & Robotics Intern | Unique World Robotics | Dubai, UAE

Mar 2026 – Apr 2026

- Built and delivered AI solutions across multiple domains in a fast-paced startup environment — rapid prototyping, iterative delivery, and cross-functional collaboration with engineers and stakeholders

Data Analyst Intern | Miles Education | Bangalore, India

Apr 2024 – Jun 2024

- Designed and deployed Power BI dashboards for KPI tracking; cleaned and transformed structured datasets to support business reporting and data-driven decision-making

EDUCATION

B.Tech Computer Science — AI & Machine Learning

2022 – 2026

JAIN University, Bangalore | CGPA: 7.8 / 10.0

CERTIFICATIONS

- Introduction to TensorFlow for AI, ML & Deep Learning — Coursera (Oct 2024)
- Fundamentals of Reinforcement Learning — Coursera (Oct 2024)
- A Deep Understanding of Deep Learning (with Python) — Feb 2024
- The Complete Machine Learning Course with Python — Sep 2023